

Fundamentals of SQL

Database Design and Presentation Final Project – Fall 2025

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Project Intro

Topic:

For this project, I chose to design a database for
a fictional small business--
“WIDGETS of CONSEQUENCE, Inc.”

Project Intro

Reason:

In real life, I am the owner of the web domain “widgetsofconsequence.com,” which I purchased with the intention of creating a satirical/comedic website about a fictional business called ‘Widgets of Consequence, Inc.’

Project Intro

Purpose:

The database I designed was conceived as both an exercise in imagination to imagine the personnel and company structure of Widgets of Consequence, as well as a preliminary exercise towards putting together a website where the database could be utilized to inform the design and world-building of the site. I envision the website to be a place where I can both have fun as well as showcase my skills and sensibilities for potential employers and clients.

Project Intro

But why “Widgets of Consequence”?

While working on an assignment for my Intro to Python class this summer, I wrote a random email address generator program, and one of the addresses the program generated struck me as funny. I had generated the names, domain names, and TLD endings from my own brain without much thought, but once I saw the address that the program had randomly generated from my stream-of-consciousness brainstorming, a whole world sprung into my mind. I eventually registered the domain name, set up e-mail addresses for the fictional founders of the business, and so it all began...

Database Design

widgetsofconsequence_db

The database I designed is comprised of 5 tables: **personnel**, **departments**, **positions**, **product_lines**, and **price_list**. Each table has a Primary Key column, and at least one Foreign Key column that refers to another table in the database. The database is intended to represent a simplified version of a small company's organizational structure and products.

Database Design

Entity Relationship Diagram:

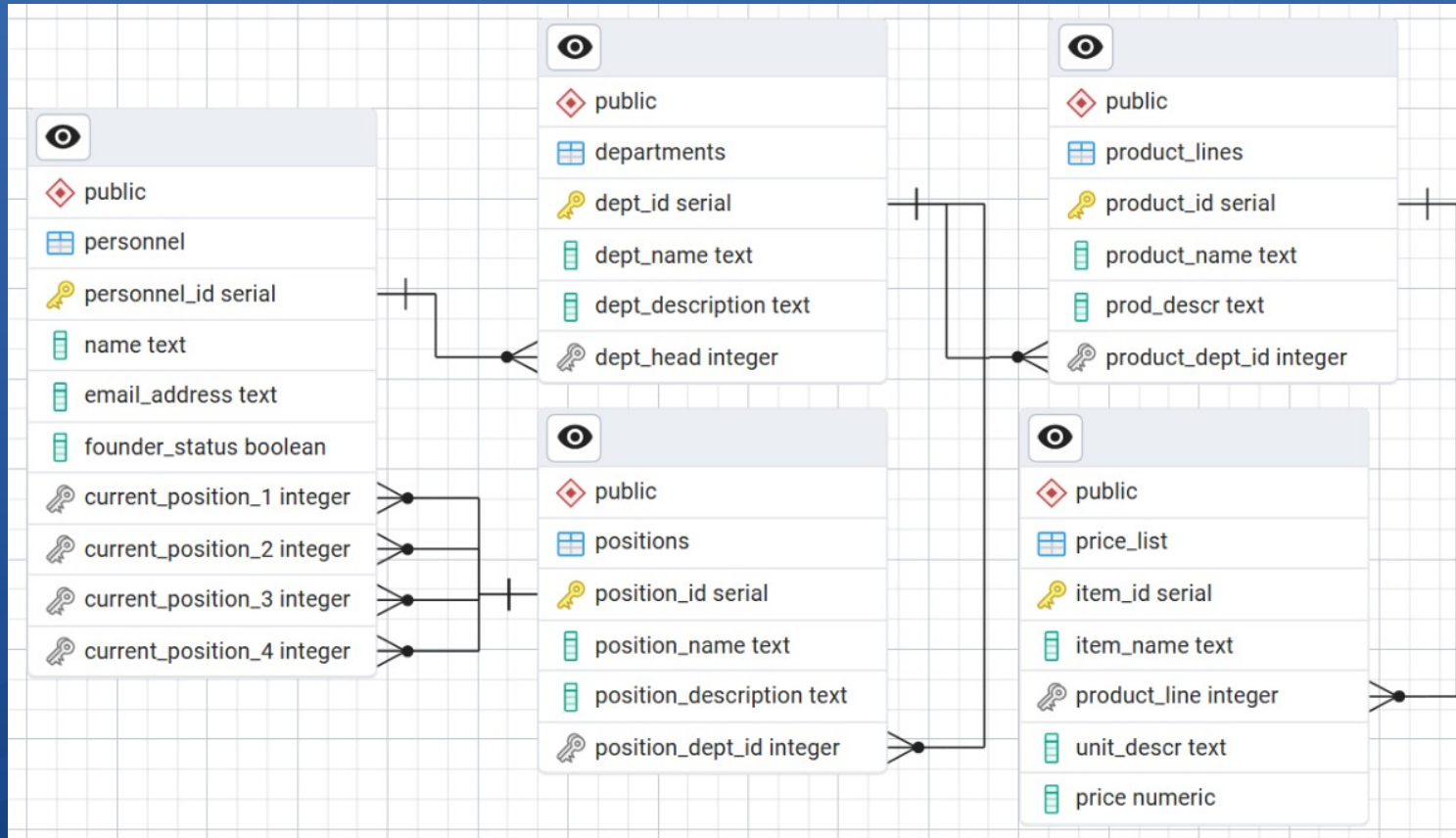


Table Briefs

personnel

The personnel table contains 8 columns: **personnel_id**, **name**, **email_address**, **founder_status**, and **current_position_1** through **current_position_4**. This table contains records for all personnel in the company, their email address, whether they are a company founder or not, and what current positions they hold. The current position columns are Foreign Keys to the **positions** table to ensure their data refers to positions in the positions table. There are multiple columns for current_positions to reflect that each person may hold multiple positions at one time.

Table Briefs

personnel

personnel_id [PK] integer	name text	email_address text	founder_status boolean	current_position_1 integer	current_position_2 integer	current_position_3 integer	current_position_4 integer
1	Wrigley Jenkins	wrigleyjenkins@widgetsofconsequence.com	true	2	9	8	1
2	Railroad Bill	railroadbill@widgetsofconsequence.com	true	4	15	9	1
3	Travis Nelson	travisnelson@widgetsofconsequence.com	true	7	17	18	1
4	The Pizza Boy	pizzaboy@widgetsofconsequence.com	true	1	10	18	1
5	Rosie Rivetts	rosierivetts@widgetsofconsequence.com	false	3	15	9	1
6	Parsnip McGee	parsnipmcgee@widgetsofconsequence.com	false	11	19	[null]	[null]
7	Delicious Aloysius	deliciousalloysius@widgetsofconsequence.com	false	14	12	9	[null]
8	Pat	pat@widgetsofconsequence.com	false	16	5	[null]	[null]
9	Diamante Pattinson	diamantepattinson@widgetsofconsequence.com	false	13	12	[null]	[null]
10	Francine Pants	francinepants@widgetsofconsequence.com	false	6	12	[null]	[null]
11	Barb Jimenez	barbjimenez@widgetsofconsequence.com	false	15	20	[null]	[null]
12	Mitch Margolin	mitchmargolin@widgetsofconsequence.com	false	15	18	[null]	[null]
13	Pinky Barbarelli	pinkybarbarelli@widgetsofconsequence.com	false	19	20	[null]	[null]

Table Briefs

positions

The positions table contains 4 columns: **position_id**, **position_name**, **position_description**, and **position_dept_id**. This table contains records for the various positions at the company, including their name and description, and which department each position is located in. The position department id column is a Foreign Key to the **department** table to ensure that each position is located within an eligible department listed in the departments table.

Table Briefs

positions

position_id [PK] integer	position_name text	position_description text	position_dept_id integer
1	Pizza Boy	Delivers the Pizza	2
2	CEO	Chief Executive Officer	1
3	Factory Foreperson	Manages factory operations and personnel	3
4	Old Tyme Widgetry Wizard	Head of all railroad-era and old tyme widgetry	6
5	Seltzer Quartermaster	Ensures consistent company-wide access to bubble water	2
6	Artists Guild Guildmaster	Wrangles all the talented painters and craftspeople in the factory artists guild	3
7	Bits & Bytes Primary Product Manager	Digital Services department head	4
8	Wiggler in Chief	Wiggling and jazzercise personal well-being lead	2
9	Widget Imagineer	Member of the R&D Widget Imagination sub-committee	7
10	COOL	Chief Operations Officer & Lead	9
11	Chief Notary	Supplier of Official Stamp of Budget Approval	8
12	Painter of Dreams	Fine Arts Oil Painter of Bespoke Consequential Widgets	3
13	Artists Guild Assistant Guildmaster	Assists Artists Guild Guildmaster	3
14	Digital to Tangible Transformations Lead	Oversees the NFT product pipeline	5
15	Factory Craftsperson	hands-on woodworking, 3-D printing, riveting, etc. in the factory	3
16	System Administrator	Keeping everything running smoothly in IT land	9
17	Webmaster	Keeping that website going	9
18	IT Backup Tech	On-call IT personnel whose main job lies elsewhere in the company, but can be called in when things get dicey	9
19	Numbers Person	Book-keeper and Accountant	8
20	Legal Eagle	Very knowledgeable certified paralegal (who also technically is a lawyer but prefers to keep that on the DL)	8
21	Founders Circle Member	Has a seat and a vote in the Founders Circle	1

Table Briefs

departments

The departments table contains 4 columns: **dept_id**, **dept_name**, **dept_description**, and **dept_head**. This table contains records for the different departments of the company—helping flesh out the shape of the company structure—including department name and description, and who the current department head is. The department head column is a Foreign Key to the **personnel** table to ensure department heads can only be selected from the list of company personnel in the personnel table.

Table Briefs

departments

dept_id [PK] integer	dept_name text	dept_description text	dept_head integer
1	Founders Circle	Highest level decision-making body at WoC	1
2	Refreshment & Relaxation	Responsible for keeping up personnel morale through keeping the snack bar, kitchen, and spa room in tip-top shape	4
3	Production	Overseeing operations at our artisanal widget factory	5
4	Digital Services	Overseeing our digital offerings, products, and services	3
5	NFTery	Cultivating our line of bespoke IRL NFTs	7
6	Planned Obsolescence	Overseeing our line of full-spectrum electronic device funeral services	2
7	Research & Development	Responsible for envisioning the next generation of consequential widgets	1
8	Finance & Accounting	Oversees the Numbers and the Money	6
9	Operations	Responsible for physical and IT infrastructure, including office supplies and networking	8

Table Briefs

product_lines

The product_lines table contains 4 columns: **product_id**, **product_name**, **prod_descr**, and **product_dept_id**. This table contains records for the different product lines at Widgets of Consequence, including name and description, and which department each under. The product department id column is a Foreign Key to the **departments** table to ensure that each product line's department can only be selected from the list of options in the departments table. (This table was originally combined with price_list table, but was separated out in the process of doing 3rd Normal Form normalization).

Table Briefs

product_lines

product_id [PK]	product_name	prod_descr	product_dept_id
1	Fridge Art Remix Keepsake Paintings	Send us a piece of art made by your child, and we will turn it into an unforgettable memory. We will first scan it into our systems and then process it through our custom generative AI LLM art bot -- the Widgeroo 5000 -- to enhance it in every way. Once our team is satisfied that the stick figure drawn by your child is truly a Widgets of Consequence masterpiece, our guild of highly trained oil painting masters (humans all), will painstakingly handpaint their interpretation of the Widgeroo 500 enhanced version of your original fridge art, and our woodworking artisans will handcraft the perfect frame to show off your new keepsake before shipping it all back to you before Christmas (or the holiday of your choice).	3
2	Real Boy NFTs	Inspired by the timeless desire of Pinocchio to become a Real Boy, our visionary wish fulfillment Imagineers are here bring you the ultimate in Non-Fungability: cold, hard physical reality! Bring your NFT to full 3-dimensional sold-state physical life by having our dedicated NFTery department make a truly one-of-a-kind sculpture out of your proprietary NFT investments. Working with wood, metal, wire, and bespoke, in-house-designed and manufactured circuit boards as well as old school animation technologies for your animated NFTs (like flipbook tech!), we will make your wishes come true! Start your Real Boy NFT collection today! (Note: must demonstrate full and legal ownership of your NFT in order to access this service).	5
3	Block/Chain IRL NFT Sculptures	Question: what is more Non-fungible than proof of ownership on the Blockchain?? Answer: actual blocks and actual chains! Instead of transforming your digital NFT to physical form, why not come at it from the other direction and have our incredible team of artisans create a one-of-a-kind sculpture for you using the power of the human mind and our advanced prompt randomizers--vintage, half-filled-in Mad Libs booklets from the 90s?? Completely Disrupt the Crypto status quo by going way way old school with your high-risk investments! Just send us a noun, a verb ending in ing, an adjective, another noun, and \$5000, and we will send you a unique, never-to-be-repeated IRL NFT Sculpture made out of a big wooden block, thick metal chain, and a custom-painted scene on each face--try hacking that! (choose from Basic--oak block + cast iron chain, and Leet--black walnut block + chrome-plated titanium chain)	5
4	Dead Phone Memorials	Planned Obsolescence got you down? Turn the creeping bug of intentionally sub-par electronic devices into a stately and beautiful feature with our comprehensive line of Dead Phone Memorials. Featuring full-spectrum phone funereal services, including ecologically friendly cremation (we remove all of the toxic bits of your phone and then crush the remainder into a pleasing memorial of your choice--choose from Classic (Your phones ashes placed in a tasteful urn), Techy (the ashes of your bricked phone get mixed in with our artisanal bricklayer mix and formed into a literal brick), and Charming (we 3-D print a miniature replica of your phone as a charm bracelet charm using your phones ashes as the substrate for the printer--charm bracelet available to purchase separately)); taxidermy (we remove and responsibly recycle the innards of your phone, fill the inside with archival-quality taxidermy resin, and then mount phone on a charming display stand complete with hand-painted, life-like screen scene and brass plaque listing Model name, Nickname, Cause and Date of Device Demise); and more options upon request--just inquire with our caring and experienced Planned Obsolescence team	6

Table Briefs

price_list

The price_list table contains 5 columns: **item_id**, **item_name**, **product_line**, **unit_descr**, and **price**. This table contains records for the specific items + their options available to purchase from Widgets of Consequence, including item name, unit description, and price; and which product line each is a part of. The product line column is a Foreign Key to the **product_lines** table to ensure that each item must be linked to a legitimate product line from the product_lines table.

Table Briefs

price_list

item_id [PK] integer	item_name text	product_line integer	unit_descr text	price numeric
1	Fridge Art Remix Keepsake Painting - SMALL	1	1 - 8x10 inch custom framed oil painting	800
2	Fridge Art Remix Keepsake Painting - MEDIUM	1	1 - 11x17 inch custom framed oil painting	1200
3	Fridge Art Remix Keepsake Painting - LARGE	1	1 - 3x5 foot custom framed oil painting	5000
4	Real Boy NFT - Static Image - SMALL	2	1 - 2x3x2 inch sculpture	850
5	Real Boy NFT - Static Image - LARGE	2	1 - 4x6x2 inch sculpture	1000
6	Real Boy NFT - Animated Image - SMALL	2	1 - 2x3x2 inch sculpture	1250
7	Real Boy NFT - Animated Image - LARGE	2	1 - 4x6x2 inch sculpture	1500
8	Block/Chain IRL NFT - BASIC	3	1 - custom 4x4x4 inch oak/iron sculpture	3500
9	Block/Chain IRL NFT - LEET	3	1 - custom 4x4x4 inch walnut/titanium sculpture	5000
10	Dead Phone Memorial - CLASSIC CREMATION	4	1 - ashes-filled urn	650
11	Dead Phone Memorial - TECHY CREMATION	4	1 - standard-sized brick	500
12	Dead Phone Memorial - CHARMING CREMATION	4	1 - bracelet charm	750
13	Dead Phone Memorial - CHARM BRACELET	4	1 - 18k gold charm bracelet	550
14	Dead Phone Memorial - TAXIDERMY	4	1 - taxidermied device w/ stand	1100

Query Briefs

QUERY #1 - WHERE

For my first query, I wanted to display a list of all IT personnel. IT is not its own department, so I first created a view which included a multi-line WHERE clause in order to capture all the possible locations of IT positions in the personnel table. This was necessary because there were three possible columns that could have contained one of the IT positions—COOL, System Admin, Webmaster, IT Backup Tech (current_position_4 is reserved for the Founders Circle, so I didn't look there). Once the view was created, I simply queried it and displayed the IT personnel.

Query Briefs

QUERY #1 - WHERE

Query

```
CREATE VIEW vw_it_personnel AS
SELECT name
FROM personnel
WHERE current_position_1 = 10
OR current_position_1 > 15
AND current_position_1 < 19
OR current_position_2 = 10
OR current_position_2 > 15
AND current_position_2 < 19
OR current_position_3 = 10
OR current_position_3 > 15
AND current_position_3 < 19;

SELECT name AS it_personnel
FROM vw_it_personnel;
```

Result

	it_personnel text 
1	Travis Nelson
2	The Pizza Boy
3	Pat
4	Mitch Margolin

Query Briefs

QUERY #2 - WHERE

For my second query, I wanted to display all personnel who were founders of the company and their e-mail address. To do this, I selected the name and e-mail address columns in the personnel table, and created a WHERE clause where **founder_status** was equal to True. That then resulted in the desired list of Widgets of Consequence company founders.

Query Briefs

QUERY #2 - WHERE

Query

```
SELECT name AS founder, email_address  
FROM personnel  
WHERE founder_status = True;
```

Result

	founder text	email_address text
1	Wrigley Jenkins	wrigleyjenkins@widgetsofconsequence.com
2	Railroad Bill	railroadbill@widgetsofconsequence.com
3	Travis Nelson	travisnelson@widgetsofconsequence.com
4	The Pizza Boy	pizzaboy@widgetsofconsequence.com

Query Briefs

QUERY #3 - JOIN

For my third query, I wanted to display a list of all departments with their corresponding department head. Since the departments table only contained an ID number for the department heads (a Foreign Key for the personnel table's Primary Key), I made a JOIN statement to bring over the name column from **personnel**, and the result successfully displayed each department's name and the corresponding department head's name.

Query Briefs

QUERY #3 - JOIN

Query

```
SELECT dept_name, name AS dept_head
FROM departments
LEFT JOIN personnel ON departments.dept_head = personnel.personnel_id
ORDER BY dept_id;
```

Result

	dept_name text	dept_head text
1	Founders Circle	Wrigley Jenkins
2	Refreshment & Relaxati...	The Pizza Boy
3	Production	Rosie Rivetts
4	Digital Services	Travis Nelson
5	NFTery	Delicious Aloysius
6	Planned Obsolescence	Railroad Bill
7	Research & Development	Wrigley Jenkins
8	Finance & Accounting	Parsnip McGee
9	Operations	Pat

Query Briefs

QUERY #4 – GROUP BY

For my fourth query, I wanted to get a count of how many distinct roles each department contained. Since some positions can be held by multiple personnel (for instance IT Backup Tech), this wasn't the same thing as getting a total head count per department. To do this, I created a view from **positions**, using the COUNT() aggregate function along with a GROUP BY statement to get a count of the distinct roles per department. I then SELECTed the view and joined it to **departments** so that I could include the department name in the query result.

Query Briefs

QUERY #4 – GROUP BY

Query

```
CREATE VIEW vw_dept_positions_count AS
SELECT position_dept_id AS dept_id, COUNT(position_name) AS distinct_roles_in_dept
FROM positions
GROUP BY position_dept_id
ORDER BY distinct_roles_in_dept DESC;

SELECT vw_count.dept_id, dept_name, distinct_roles_in_dept
FROM vw_dept_positions_count AS vw_count
LEFT JOIN departments ON vw_count.dept_id = departments.dept_id
ORDER BY distinct_roles_in_dept DESC;
```

Result

	dept_id integer	dept_name text	distinct_roles_in_dept bigint
1	3	Production	5
2	9	Operations	4
3	8	Finance & Accounting	3
4	2	Refreshment & Relaxati...	3
5	1	Founders Circle	2
6	6	Planned Obsolescence	1
7	7	Research & Development	1
8	5	NFTery	1
9	4	Digital Services	1

CONCLUSION

Thank you for viewing my presentation.

(In case you were wondering, the e-mail address that set the whole 'Widgets of Consequence' thing off was pizzaboy@widgetsofconsequence.com).

Note: All screenshots taken from PgAdmin4 except for Slide 20, which was a simulated screenshot made in Libre Office Calc in order to use its text wrap function for the extra-long text.